

# Amul Chourasia

PHD STUDENT

Rupnagar, Punjab

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## Education

### Indian Institue of Technology Ropar

PHD IN COMPUTER SCIENCE & ENGINEERING

Rupnagar, Punjab, India

January 2025 - Present

### Indian Institue of Information Technology Guwahati

MTech IN COMPUTER SCIENCE & ENGINEERING, 8.05/10.00

Guwahati, Assam, India

August 2021 - May 2023

### University Institute of Technology, Barkatullah University

BACHELOR IN ENGINEERING, 7.63/10.00

Bhopal, India

2016-2020

### Shri Ram H.S. School

HIGHER SECONDARY EDUCATION (CLASS 12), 81.6%

Madhya Pradesh, India

2014-2016

### Kendriya Vidyalaya Seoni

CLASS 10, 7.00/10.00

Madhya Pradesh, India

2014

## Publication

### IEEE Region 10 Conference (TENCON 2023)

A CHOURASIA, V NAOSEKHAM AND N SAHU. ATTENTION-FUSED SHALLOW NETWORK FOR UNDERWATER OBJECT DETECTION

Chiang Mai, Thailand

October - November 2023

- Developed a two-stage method utilizing PCA-fusion for image enhancement and an attention-fused lightweight model for object detection, achieving superior performance on the URPC2019 dataset
- DOI: <https://ieeexplore.ieee.org/document/10322535>

### Computer Analysis of Images and Patterns: 20th International Conference (CAIP 2023)

V NAOSEKHAM, M ISLAM, A CHOURASIA AND N SAHU. EMBIL: AN ENGLISH-MANIPURI BI-LINGUAL BENCHMARK FOR SCENE

Limassol, Cyprus

September 2023

TEXT DETECTION AND LANGUAGE IDENTIFICATION

- Developed EMBIL, a scene image dataset with English and Manipuri text, and proposed SceneTextYOLO-Net based on YOLOv5 for simultaneous text detection and language identification, achieving promising results on EMBIL and ICDAR2015 datasets
- DOI: [https://doi.org/10.1007/978-3-031-44237-7\\_7](https://doi.org/10.1007/978-3-031-44237-7_7)

## Work Experience

### Assistant Professor

INSTITUTE OF TECHNICAL EDUCATION AND RESEARCH (ITER), SOA

Bhubaneswar, India

June 2023 - December 2024

- Taught the Introduction to Computer Programming using Java course for 1st sem students, fostering a strong foundational understanding of object-oriented programming
- Guided 2nd sem students through Data Structures and Algorithms in Java, emphasizing problem-solving techniques, algorithm optimization, and efficient data handling strategies
- Instructed in FDP "Python Programming From Modular Basics to Advanced Techniques" training on Python programming, covering both foundational and advanced techniques
- Played a pivotal role as part of the evaluator panel, providing insightful feedback and mentoring participants in the development and innovation process during the high-stakes 24-hour Institute level Hackathon competition
- Teaching Application Development Program using Spring and Spring Boot for 7th-semester students, focusing on building scalable, enterprise-level applications

### Research Intern

INDIAN INSTITUTE OF INFORMATION TECHNOLOGY GUWAHATI

Guwahati, Assam, India

May 2022 - July 2022

- Conducted extensive research for the survey segment by reviewing ten academic papers, preparing comprehensive reports on the topic "Detecting COVID-19 using Deep Learning"
- Collaborated on the utilization of CNN deep learning models such as VGG-16 and ResNet

## UI/UX Designer Intern

THE DAIRING COMPANY

Guwahati, Assam, India

January 2022 - April 2022

- Contributed to designing and delivering user-friendly online UI/UX widget components for internal teams, utilizing HTML, CSS, JavaScript, and jQuery to ensure seamless integration into various web pages and applications
- Leveraged modern front-end technologies such as Ember.js, HTML, and jQuery to significantly improve product usability and streamline the user experience, making applications more intuitive and efficient for end-users

## Projects

### Vehicle Number Plate Detection

COURSE: TOPICS IN AI

Rupnagar, Punjab

October 2025

- Developed a robust license plate detection framework integrating multi aspect feature refinement to handle challenging visual conditions.

### Crowd Face Detection

COURSE: TOPICS IN AI

Rupnagar, Punjab

November 2025

- Implemented state of the art methods to develop a robust framework

### Diabetes Macular Oedema detection from OCT-scanned images

ONGOING COLLABORATIVE PROJECT

Bhubaneswar, Odisha

May 2025 - Present

### IoT Gas Monitoring System

COURSE: IOT

Guwahati, Assam

March 2023

- Utilized NodeMCU, MQ2 gas sensor, and a buzzer to develop a gas monitoring system that detects and alerts on gas levels
- Implemented MQTT for real-time data transmission and used ThingSpeak for cloud-based data analysis and monitoring

### Online Feedback System

COURSE: DBMS

Guwahati, Assam

November 2021

- Created an intuitive user interface using HTML, CSS, and JavaScript for seamless student feedback submission
- Developed the server-side logic and database management using PHP and MySQL to efficiently handle and store feedback data

## Certifications

- Machine learning and AI by working on Hands-on Projects
- Java Programming for Complete Beginner
- 100 Python Beginners Coding Practices
- Go from Beginner to Advanced with Google Docs
- C++ Programming Step By Step From Beginner To Ultimate Level
- Learn Figma for Web Design, User Interface, UI UX
- Canva 2022 Graphic Design Class | Latest Canva Version 3.0
- Python Programming For absolute beginners : Hands-on Python
- Complete Drawing Megacourse: Beginner to Expert
- C++ programming through C

## Skills

PROGRAMMING LANGUAGE, FRAMEWORKS, DATABASE, TOOLS

- Programming Language: Java, Python, C, C++, HTML, CSS | Frameworks: OpenCv, Bootstrap, Spring, SpringBoot
- Database: MySQL | Design Tools: Figma, Canva, Github

## Activity

- Volunteered in International Conference on Computer Vision and Image Processing (CVIP 2025) in December 2025.
- Volunteered in ACM India Winter School 2025 December.
- 3 star coder in HackerRank
- Served as a Publicity Chair and Organizing Committee Member for the International Conference on Advancements in Artificial Intelligence and Machine Learning for Security (ICAAIMLS-2024), organized by ITER, SOA
- Instructed in FDP "Python Programming From Modular Basics to Advanced Techniques" in July 2024
- National Level Short-Term Training Programme on Statistical Machine Learning in August 2023
- Participated workshop on Brain Computer Interface organised by IIIT Guwahati, Dept. of Science & Technology Govt. of India
- Participated in National Level Quiz, CHIMERA-X in NIT Bhopal
- Participated in Scout & Guide Programs from 2010-2016